

# Applied Regression Ass 1

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## (a) Why is the model linear and non-linear?

We model the circle area  $A$  as

$$A = \beta_0 + \beta_1 x^2 + \varepsilon.$$

- **Linear (in parameters):** The mean function is an affine combination of  $\beta_0, \beta_1$ ; i.e., linear in the unknown coefficients.
- **Non-linear (in predictor):** The regressor is  $x^2$ , a non-linear transform of  $x$ .

## (b) Interpretations of $\beta_0$ and $\beta_1$

- $\beta_0$ : Expected area when  $x = 0$  (intercept). For true circles, we expect  $\beta_0 \approx 0$ .
- $\beta_1$ : Expected change in area **per unit increase in  $x^2$** . Geometrically, the true value is  $\pi$  since  $A = \pi x^2$ .

## (c) Why include the error term $\varepsilon$ ? Give two sources.

We include  $\varepsilon$  to capture randomness and model imperfections, e.g.: - **Measurement error** in radius or area. - **Omitted factors/model misspecification** (e.g., deformation, thickness), rounding, and sampling variability.

## (d) Sample version for $n$ observations

For  $i = 1, \dots, n$ ,

$$A_i = \beta_0 + \beta_1 x_i^2 + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

## (e) Vector/matrix form and dimensions

Let

$$\mathbf{A} = \begin{bmatrix} A_1 \\ \vdots \\ A_n \end{bmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1^2 \\ \vdots & \vdots \\ 1 & x_n^2 \end{bmatrix}_{n \times 2}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{2 \times 1}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n \times 1}.$$

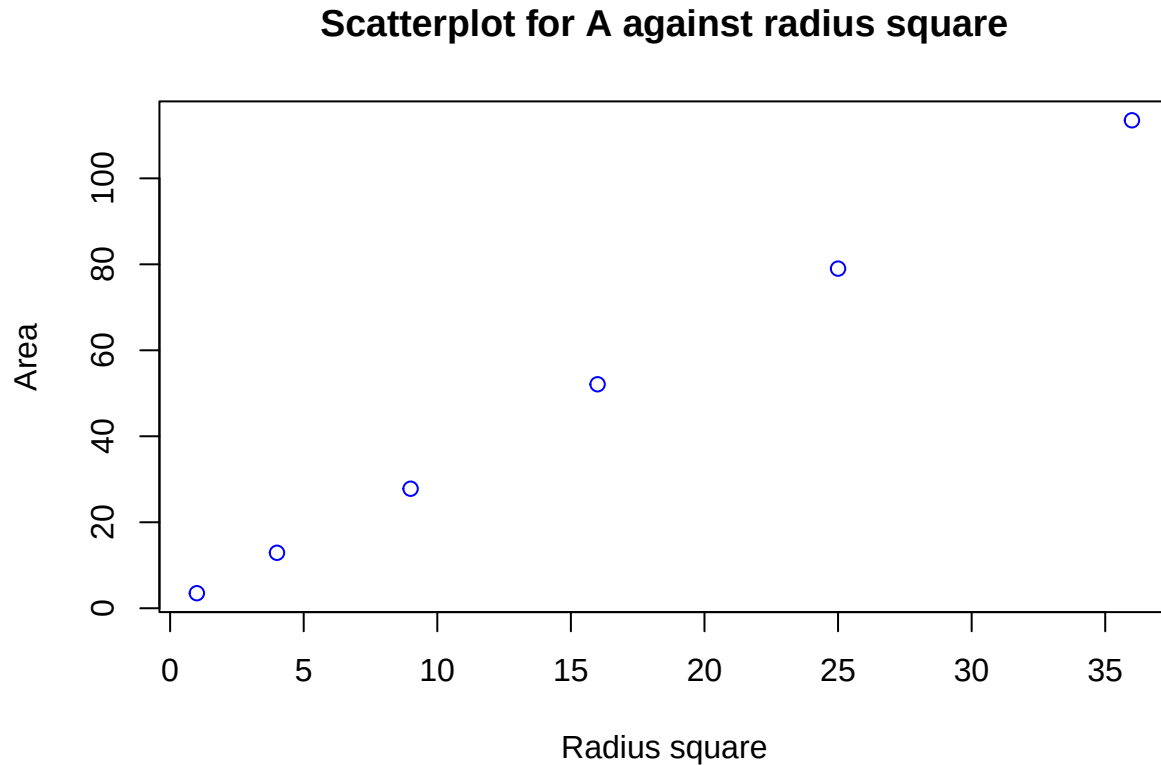
Then

$$\mathbf{A} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

**Dimensions:**  $\mathbf{A} \in \mathbb{R}^{n \times 1}$ ,  $\mathbf{X} \in \mathbb{R}^{n \times 2}$ ,  $\boldsymbol{\beta} \in \mathbb{R}^{2 \times 1}$ ,  $\boldsymbol{\varepsilon} \in \mathbb{R}^{n \times 1}$ .

(f) Use to fit the model to the data (see next slide)

(i) Graph a scatterplot for A against  $x^2$

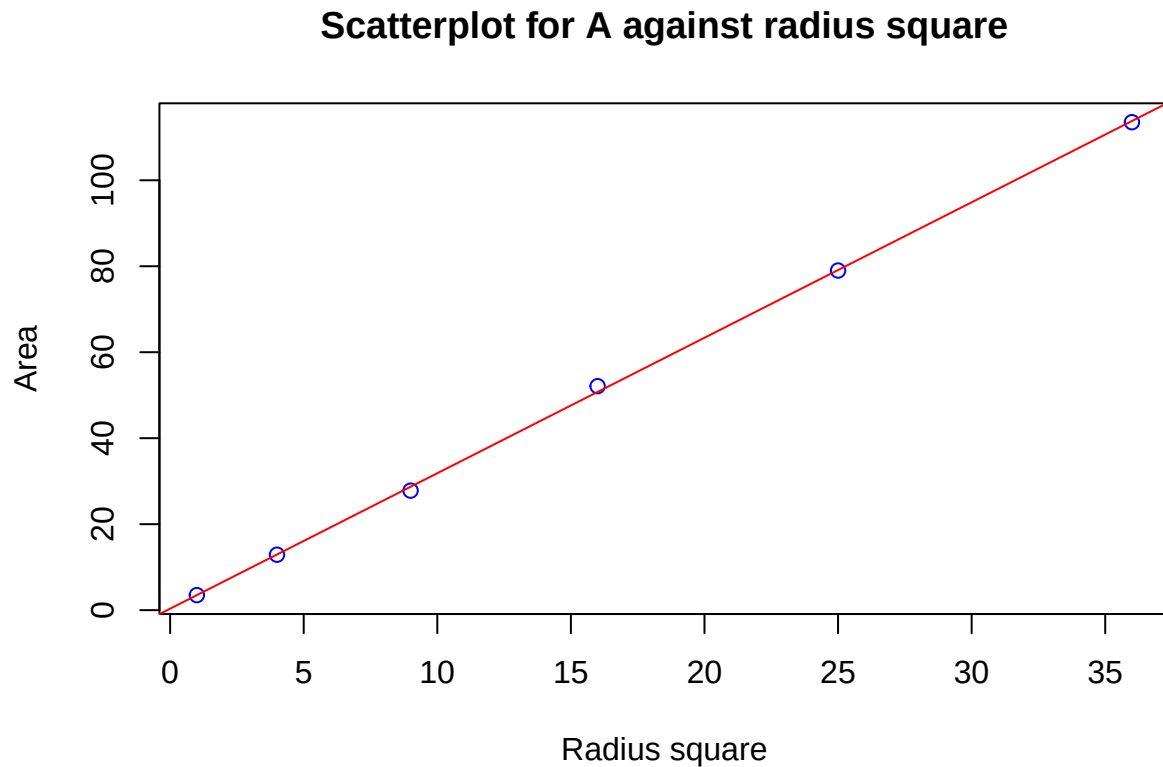


(iii) Run linear regression analysis in R for the simple linear regression model. Copy the output and write the equation for the fitted model.

```
##
## Call:
## lm(formula = area ~ radius_squared, data = our_data)
##
## Residuals:
##      1      2      3      4      5      6
## 0.008158 -0.045334 -0.901155  1.340697 -0.119780 -0.282585
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.34068    0.53539   0.636   0.559
## radius_squared  3.15116    0.02749 114.609 3.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8225 on 4 degrees of freedom
## Multiple R-squared:  0.9997, Adjusted R-squared:  0.9996
## F-statistic: 1.314e+04 on 1 and 4 DF, p-value: 3.476e-08
```

$$\hat{A} = 0.34068 + 3.15116 \cdot (\text{radius}^2)$$

(iii) Using R, graph the fitted model on the scatterplot in (i)



(g) Compare simple vs. multiple regression

We compare

- **Simple model:**  $A = \beta_0 + \beta_1 x^2 + \varepsilon$

- **Multiple model:**  $A = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

with  $x = \text{radius}$ ,  $x^2 = \text{radius\_squared}$ .

(i) Why is  $A = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$  linear and multiple?

- **Linear (in parameters):**  $A$  is an affine function of the unknowns  $\beta_0, \beta_1, \beta_2$  (no products/powers of  $\beta$ 's).
- **Multiple:** it uses **two regressors** ( $x$  and  $x^2$ ) for one response, hence a multiple regression.

(ii) Sample (observation-level) model

For  $i = 1, \dots, n$ :

$$A_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

(iii) Vector/matrix form and dimensions

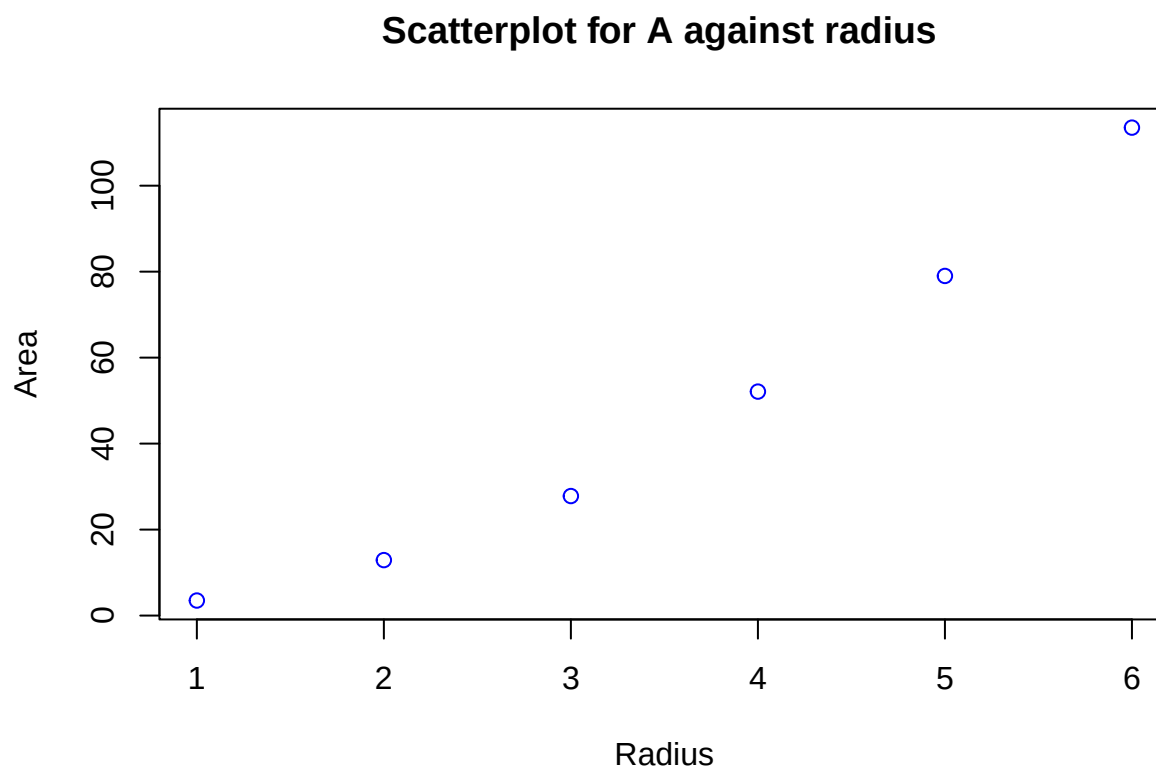
Let

$$\mathbf{A} = \begin{bmatrix} A_1 \\ \vdots \\ A_n \end{bmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}_{n \times 3}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}_{3 \times 1}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n \times 1}.$$

Then  $\mathbf{A} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ .

**Dimensions:**  $\mathbf{A} \in \mathbb{R}^{n \times 1}$ ,  $\mathbf{X} \in \mathbb{R}^{n \times 3}$ ,  $\boldsymbol{\beta} \in \mathbb{R}^{3 \times 1}$ ,  $\boldsymbol{\varepsilon} \in \mathbb{R}^{n \times 1}$ .

(iv) Graph a scatterplot for A against x



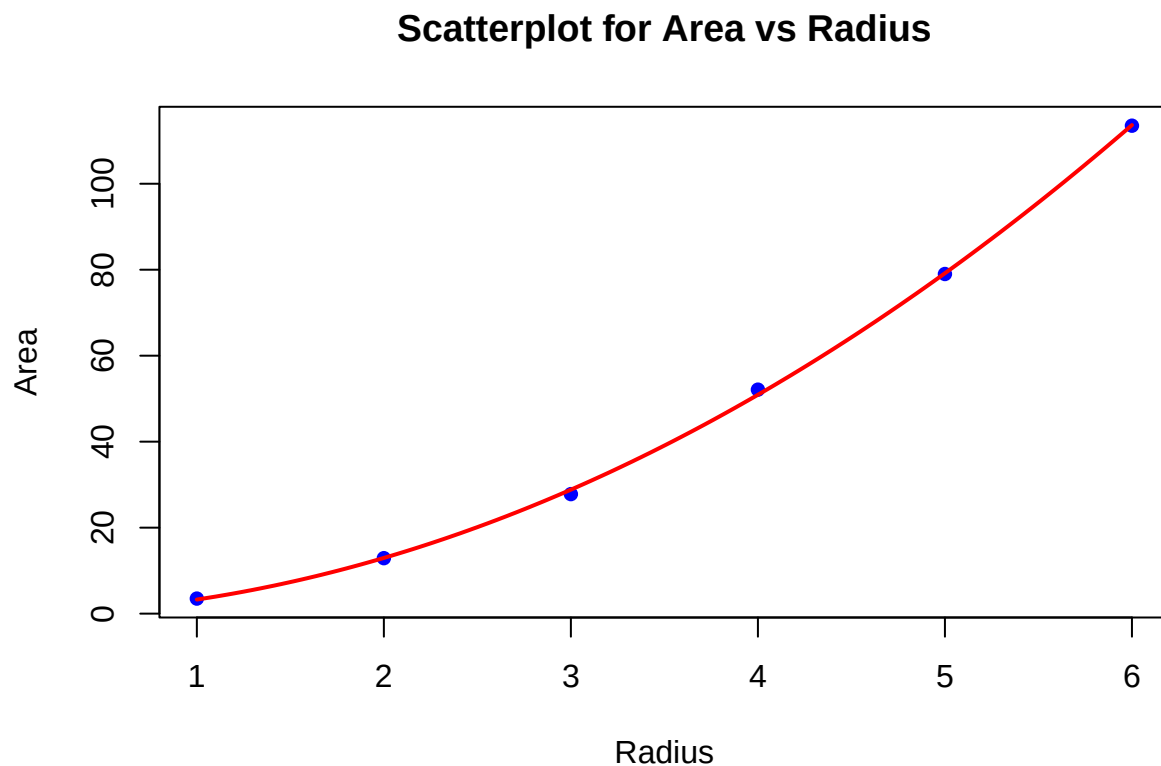
(v) Run linear regression analysis in R for the simple linear regression model. Copy the output and write the equation for the fitted model.

```
##  
## Call:
```

```
## lm(formula = area ~ radius + radius_squared, data = our_data)
##
## Residuals:
##      1      2      3      4      5      6
## 0.22500 -0.05643 -1.03429  1.19143 -0.17929 -0.14643
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.2100     1.6643   -0.126  0.90757
## radius         0.3868     1.0888    0.355  0.74592
## radius_squared  3.0982     0.1523   20.348  0.00026 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9303 on 3 degrees of freedom
## Multiple R-squared:  0.9997, Adjusted R-squared:  0.9995
## F-statistic: 5133 on 2 and 3 DF, p-value: 4.993e-06
```

$$\hat{A} = -0.2100 + 0.3868 \cdot (\text{radius}) + 3.0982 \cdot (\text{radius}^2)$$

(vi) Using R, graph the fitted model on the scatterplot in (i)



### (vii) Compare models using Adjusted $R^2$

From *Elementary Statistics*: the adjusted coefficient of determination penalizes extra predictors and is

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)},$$

where  $n$  is the sample size,  $p$  is the number of estimated parameters (including the intercept), SSE is the residual sum of squares, and SST is the total sum of squares.

We use  $\bar{R}^2$  to compare:

- **Simple model:**  $A \sim x^2$  (only `radius_squared`)
- **Multiple model:**  $A \sim x + x^2$  (`radius` and `radius_squared`)

The model with the **higher**  $\bar{R}^2$  achieves a better balance of fit and parsimony.

```
## Adjusted_R2_Simple Adjusted_R2_Multiple
## 0.9996195 0.9995131
```

#### Interpretation: Results:

- Adjusted  $R^2$  (Simple model,  $A \sim x^2$ ): **0.9996**
- Adjusted  $R^2$  (Multiple model,  $A \sim x + x^2$ ): **0.9995**

#### Interpretation:

Although both models explain almost all the variation in the data, the **simple regression model** with only  $x^2$  yields a slightly higher adjusted  $R^2$  (0.9996 vs. 0.9995).

Because adjusted  $R^2$  penalizes model complexity, this indicates that adding the linear term  $x$  does not improve model performance sufficiently to justify the extra parameter. Therefore, the **simple quadratic model** provides a more balanced fit than the multiple regression model.